



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL AIR DEFENCE

MADE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

TACTICAL AIR VEHICLE SPECIFICATIONS & OPTRONICS

NANO-SCOUT & OVERWATCH
AERIAL PLATFORM ENGINEERING

A high-performance two-tier indigenous air vehicle family engineered to operate in tactical symbiosis—pairing soldier-deployable sub-500g micro Nano-scouts for immediate obstacle overflight with long-endurance high-altitude Overwatch relays delivering continuous multi-spectral overwatch and airborne mesh gateway routing.

NANO-SCOUT (350G CLASS)

OVERWATCH VTOL (20KG CLASS)

MULTI-SPECTRAL EO/IR & LRF

SUB-ZERO HIMALAYAN RATED

NANO-SCOUT MICRO-UAS

TIER 01 • FORWARD SCOUT



SUB-500G POCKET DEPLOYABLE

ACOUSTIC <38 dB

ULTRA-LOW ACOUSTIC SQUAD SCOUT & PERIMETER PROBE

Frontline infantry recon asset hand-deployable in under 15 seconds. Designed for forward crest overflight, urban room entry, and silent perimeter breach verification with zero thermal reflection.

ALL-UP WEIGHT	350-450 grams
FLIGHT ENDURANCE	18-22 minutes
OPERATIONAL RADIUS	3-5 km Line-of-Sight
DASH SPEED	45 km/h max
SENSOR SUITE	HD Daylight + Micro LWIR Thermal
ACOUSTIC SIGNATURE	<38 dB @ 25 meters
DEPLOYMENT MODE	Hand / Pocket Launch (<15s)

OVERWATCH VTOL RELAY

TIER 02 • AIRBORNE RELAY



~20KG CLASS HYBRID VTOL

8,000+ FT SERVICE CEILING

HIGH-ALTITUDE AIRBORNE RELAY & PERSISTENT ISTAR MOTHERSHIP

Runway-free tactical VTOL operating as the airborne command-and-control node. Bridges micro-swarm video feeds to tactical headquarters while executing 24/7 high-magnification electro-optical reconnaissance.

MAX TAKEOFF WEIGHT	~20 kg Class
MISSION ENDURANCE	120-180 minutes
OPERATIONAL RADIUS	100+ km BVLOS
SERVICE CEILING	8,000+ ft AMSL
GIMBAL OPTRONICS	30x Optical Zoom + Radiometric FLIR
LASER RANGEFINDER	Class 1 Eye-Safe (±1m @ 3km)
AVIONICS ARCHITECTURE	Dual-Band Anti-Jam NAVIC + Mesh

INTEGRATED OPTRONICS & TARGETING PAYLOADS

MULTI-SPECTRAL EO/IR & LASER TELEMETRY

Equipped with gyro-stabilized multi-sensor gimbals engineered for 24/7 all-weather surveillance across contested borders. Seamlessly switches between high-magnification visible spectrum optics and thermal thermography to defeat optical camouflage, mist, and night darkness.

DAYLIGHT 30X ZOOM

Sony Low-Light Sensor

High-definition target identification at 3,500m stand-off with automated motion tracking.

640x512 LWIR THERMAL

Uncooled Microbolometer

12 µm pixel pitch detects personnel heat signatures through smoke, foliage, and dense mist.

LASER RANGEFINDER

MGRS Target Geolocation

Computes instant target GPS/NAVIC grid coordinates for artillery and air defence handoff.



-20°C to +55°C

THERMAL OPERATING RANGE

Sub-zero active battery conditioning and high-desert cooling radiators.

45 km/h

MOUNTAIN GUST HANDLING

Tested against severe Himalayan wind shear and ridge downdrafts.

IP54 Enclosure

ENVIRONMENTAL PROTECTION

Sealed avionics bays resistant to driven rain, desert sand, and mountain dust.

MIL-STD-810H

STRUCTURAL RUGGEDIZATION

Carbon-composite monocoque construction built for rough tactical field handling.